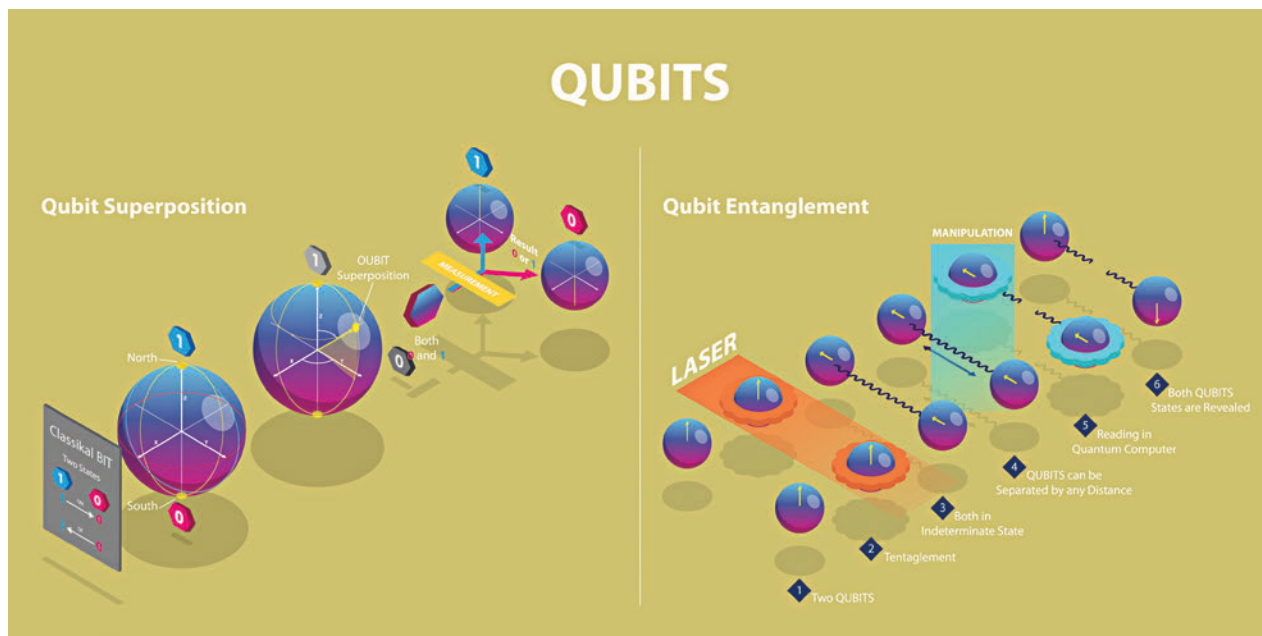


Mathematically, a qubit's state is represented as a linear combination of its basis states $|0\rangle$ and $|1\rangle$. Imagine a globe where the north pole represents $|0\rangle$, the south pole represents $|1\rangle$, and any point on the surface represents the qubit's state—a superposition of $|0\rangle$ and $|1\rangle$.

Physically, qubits can be realised using various technologies, such as trapped ions held in place by electromagnetic fields or superconducting circuits cooled to near absolute zero. These methods harness quantum phenomena to encode and process information in ways classical systems cannot.

Superposition and Entanglement

Superposition allows a qubit to be in a combination of all possible states. When we measure a qubit in superposition, we might find it in the state $|0\rangle$ with a certain probability and $|1\rangle$ with another, the outcome being inherently unpredictable.



Entanglement is a uniquely quantum phenomenon where qubits become so deeply linked that the state of one instantly influences the state of another, no matter the distance separating them. This phenomenon is not only mystifying but also a powerful resource for quantum computing, enabling complex computational tasks that classical computers struggle with.

Applications and Examples

Consider quantum cryptography, which uses entangled qubits to create secure communication channels. Any attempt to eavesdrop on the communication disturbs the entanglement, alerting the communicating parties to the intrusion. This application showcases how quantum mechanics principles can be harnessed for real-world technologies.

Quantum Gates and Circuits

Quantum gates manipulate qubits, transforming their states. They are the quantum analogs of classical logic gates but operate in a fundamentally different way, exploiting the principles of superposition and entanglement. Quantum circuits are sequences of quantum gates, designed to perform complex calculations. Building a quantum circuit involves arranging gates in a way that they apply the desired transformations to the qubits, guiding the system from an initial state to one that encodes the solution to a problem.

Common Quantum Gates

The Hadamard gate, for example, puts a qubit into a state of superposition, equally likely to be measured as $|0\rangle$ or $|1\rangle$. The CNOT gate, or controlled-NOT, entangles two qubits, flipping the second qubit (target) if the first qubit (control) is in state $|1\rangle$.

Measuring Quantum States

Measuring a qubit in superposition forces it into one of its basis states, a phenomenon that collapses the superposition. The outcome is probabilistic, dependent on the qubit's state prior to measurement. This inherent unpredictability is not a limitation but a feature, enabling quantum algorithms that, paradoxically, use probability to achieve certainty more efficiently than classical algorithms.

Implications for Computing

The act of measurement is central to quantum computing, marking the end of a computation where the quantum state's information becomes accessible to the classical world. Quantum error correction and algorithms are designed